

Ex. No 5 Use Autopsy to create a case and import evidence

Date:

Description :

Autopsy is an open-source digital forensics platform used for analyzing and extracting data from digital devices. Here's a step-by-step process on how to use Autopsy for a basic forensic investigation:

Drive link: https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v

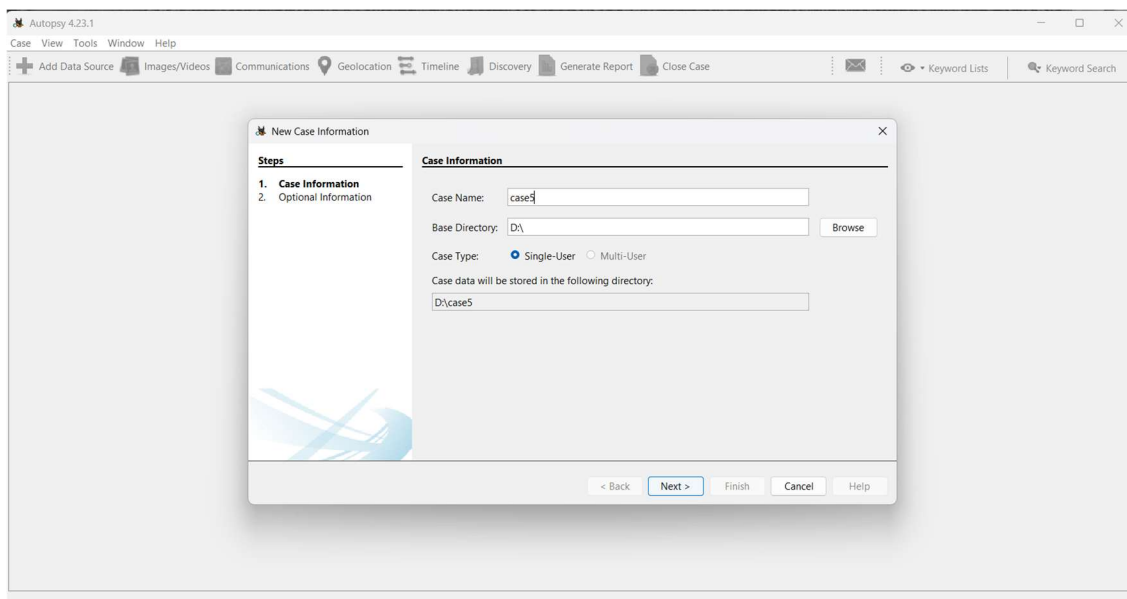
Download files: [4Dell Latitude CPi.E01](#), [4Dell Latitude CPi.E02](#)

1. Installation

- Download and Install: Autopsy can be downloaded from the official website. Follow the installation instructions based on your operating system (Windows, Linux, or macOS).

2. Starting a New Case

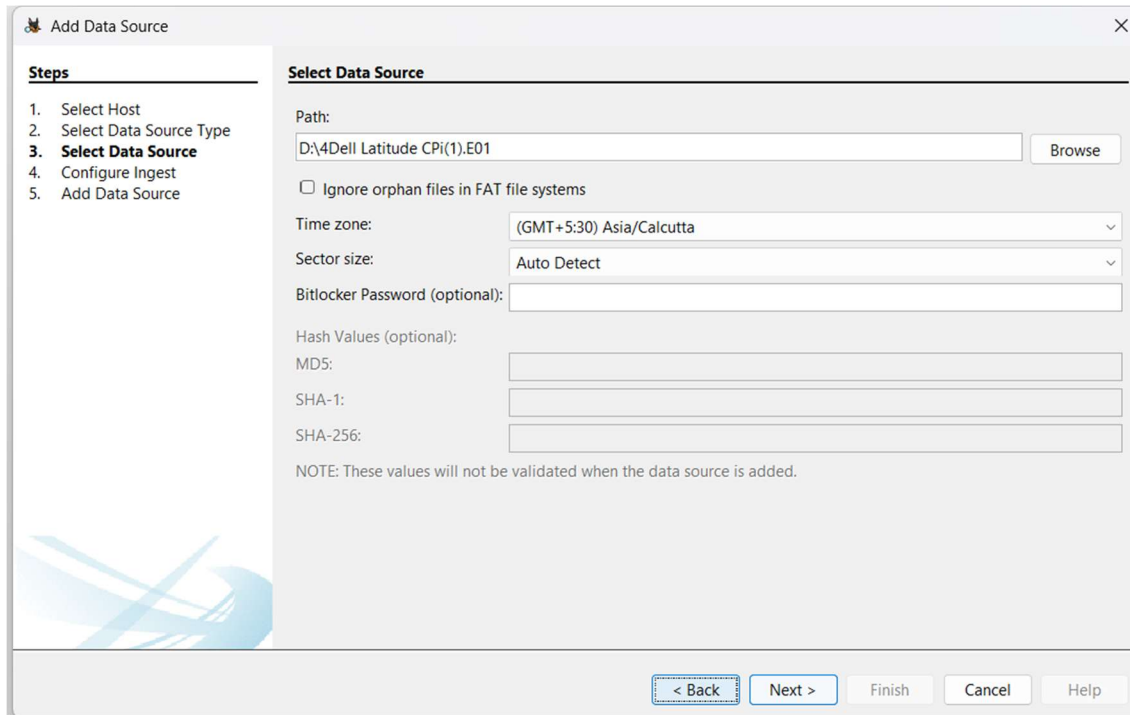
- Open Autopsy: Launch the application after installation.
- Create a New Case:
 - Click on New Case.
 - Enter the case name and location where the case data will be stored.
 - Fill in the details like the case number, examiner's name, etc., and click Next.



3. Adding a Data Source

- Choose the Type of Data Source:
 - After creating a case, you'll be prompted to add a data source.
 - You can add disk images, directories, logical files, or local disks.

- Select the Data Source:
 - Browse to the location of the image file (e.g., .E01, .dd, or .raw), physical disk, or directory you want to analyze.
 - https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v

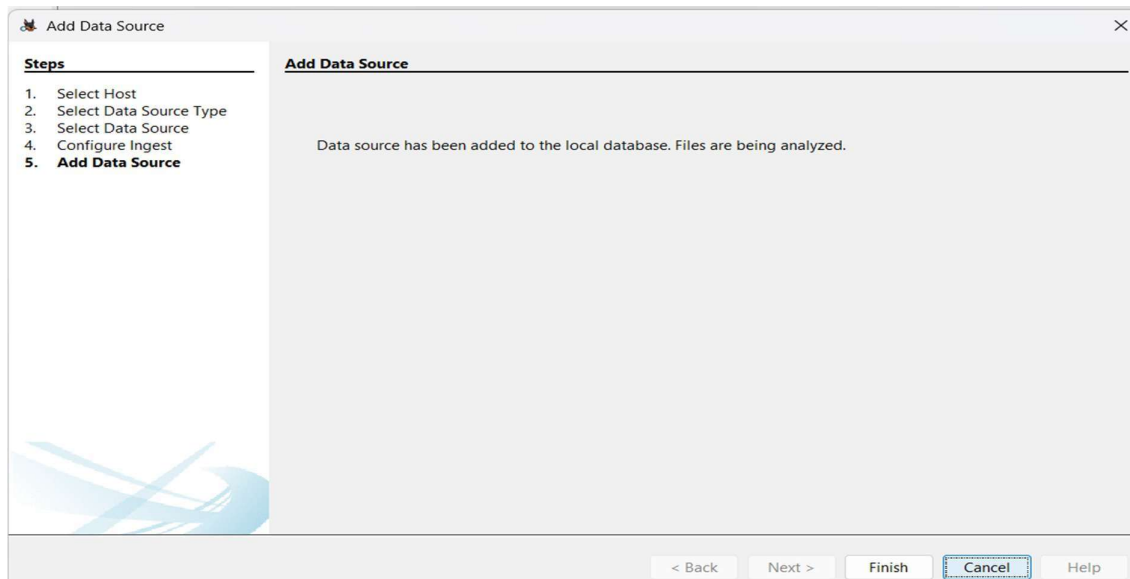


The screenshot shows the 'Add Data Source' dialog box with the 'Select Data Source' step active. The 'Steps' list on the left includes: 1. Select Host, 2. Select Data Source Type, 3. **Select Data Source**, 4. Configure Ingest, and 5. Add Data Source. The main area contains the following fields and options:

- Path:** A text box containing 'D:\4Dell Latitude Cpi(1).E01' and a 'Browse' button.
- ☐ Ignore orphan files in FAT file systems
- Time zone:** A dropdown menu set to '(GMT+5:30) Asia/Calcutta'.
- Sector size:** A dropdown menu set to 'Auto Detect'.
- Bitlocker Password (optional):** A text box.
- Hash Values (optional):** Three text boxes for MD5, SHA-1, and SHA-256.

A note at the bottom states: 'NOTE: These values will not be validated when the data source is added.' The bottom navigation bar includes buttons for '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

- Configure Ingest Modules:
 - Autopsy allows you to select specific analysis modules such as File Type Identification, Keyword Search, Hash Lookup, etc.
 - You can enable or disable these based on your investigation needs.
- Start Analysis: Click Next to start the analysis.



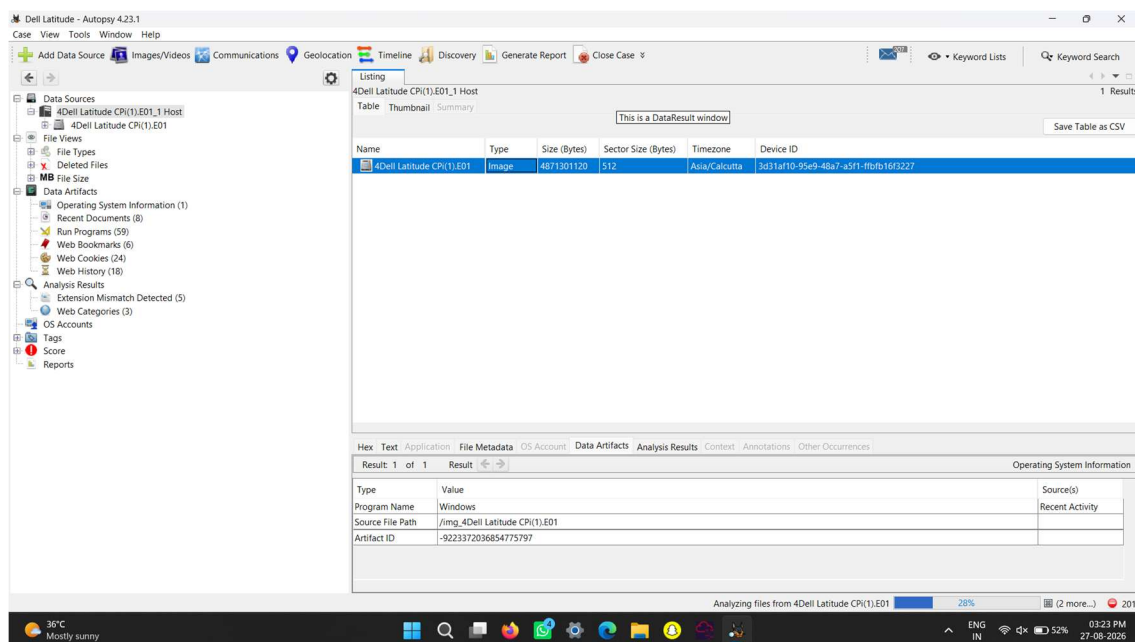
The screenshot shows the 'Add Data Source' dialog box with the 'Add Data Source' step active. The 'Steps' list on the left includes: 1. Select Host, 2. Select Data Source Type, 3. Select Data Source, 4. Configure Ingest, and 5. **Add Data Source**. The main area displays the message: 'Data source has been added to the local database. Files are being analyzed.' The bottom navigation bar includes buttons for '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

4. Initial Analysis and Overview

- Ingest Progress: As Autopsy processes the data source, you'll see the progress in the lowerleft corner.
- Explore the Resulting Artifacts:
 - Autopsy automatically categorizes findings such as web artifacts, file system metadata, and communication records.
- Use the Tree Viewer:
 - On the left pane, you'll see a tree structure where you can explore different aspects like File System, Web History, Email, etc.

5. Detailed Analysis

- Keyword Search:
 - You can perform specific keyword searches using the Keyword Search module.
 - Use pre-configured lists or enter custom keywords.
- File Analysis:
 - Navigate through files and folders under the File Types or File System section.
 - Open, view, or export files for further examination.
- Timeline Analysis:
 - Use the Timeline module to visualize events based on timestamps.
 - This can help track user activity over time.
- Hash Analysis:
 - Compare file hashes with known databases to identify known good or bad files.



6. Reporting

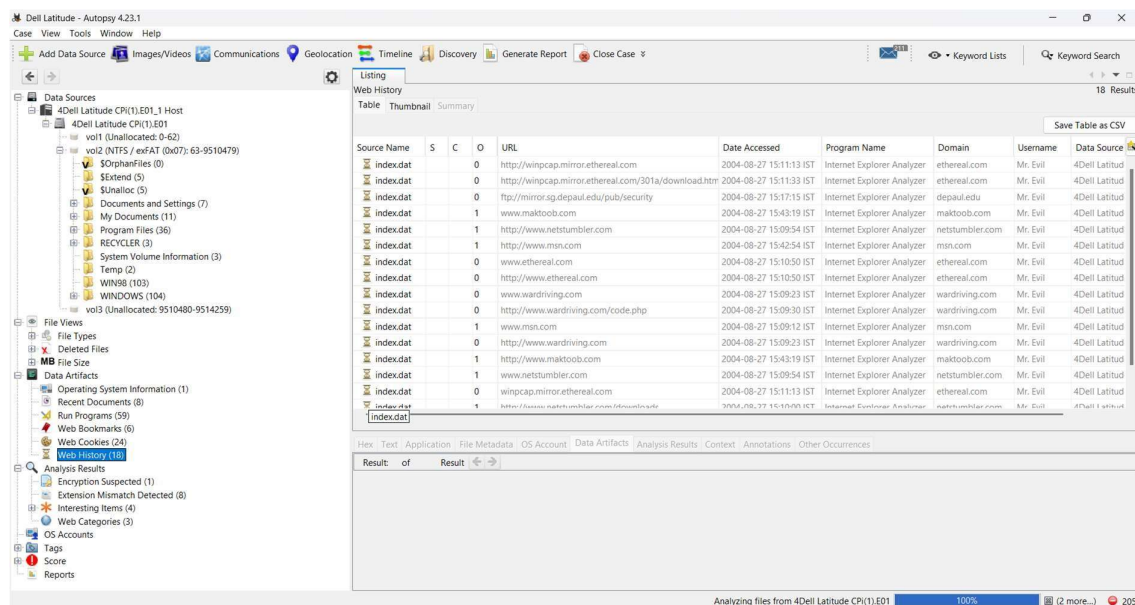
- Generate a Report:
 - After analyzing the data, click on Generate Report from the toolbar.
 - Choose the type of report (HTML, CSV, Excel, etc.).
 - Select which parts of the analysis you want to include in the report.
- Export Findings:
 - Export individual files or artifacts that you need for your report or further analysis.
- Final Review:
 - Review the report to ensure it includes all relevant information.
 - Save or print the report for use in your case.

7. Case Closure

- Close the Case:
 - Once you have completed your investigation, close the case within Autopsy.
- Archiving:
 - Ensure all data and reports are properly archived according to your organization's policies.

8. Advanced Features (Optional) • Custom Ingest Modules:

- Autopsy allows for custom modules to be added if you need specific analysis tools not included by default.
- Collaboration:
 - Autopsy can be configured for multi-user cases if you're working in a team environment.



Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

A new forensic case was successfully created in Autopsy, and the E01/E02 evidence image files were imported as the data source. The evidence was processed using the selected ingest modules, and the resulting forensic artifacts were successfully analyzed and reviewed.